

✓ 01 — Data acquisition and metadata

I query the Allen Brain Observatory Visual Coding metadata through AllenSDK. I do not include raw NWB files in Git because the dataset is large; I cache them locally only when I explicitly request download.

```
from pathlib import Path
import sys

ROOT = Path.cwd()
if ROOT.name == "notebooks":
    ROOT = ROOT.parent
if str(ROOT / "src") not in sys.path:
    sys.path.insert(0, str(ROOT / "src"))

from v1_manifold.config import load_config, get_paths, set_global_seed
from v1_manifold.visualization import set_publication_style, save_figure
from v1_manifold.utils import save_table

cfg = load_config(ROOT / "configs" / "default.yaml")
cfg["paths"]["root"] = str(ROOT)
paths = get_paths(cfg)
set_global_seed(cfg["project"]["random_seed"])
set_publication_style()
print(f"Project root: {ROOT}")
```

Project root: c:\Users\Peter\Documents\projects\NeuroAI\latent-manifold-v1-na

```
from v1_manifold.data_access import get_boc, query_v1_natural_movie_experiment

boc = get_boc(paths.allen_manifest)
experiments = query_v1_natural_movie_experiments(boc, cfg)
metadata_path = save_experiment_catalog(experiments, paths.external_metadata)
print(f"Saved {len(experiments)} selected experiments to {metadata_path}")
experiments.head()
```

```
c:\Users\Peter\.neuro\Lib\site-packages\allensdk\core\brain_observatory_nwb_d
from pkg_resources import parse_version
Saved 3 selected experiments to C:\Users\Peter\Documents\projects\NeuroAI\lat
      id imaging_depth targeted_structure cre_line reporter_line acq
```

0	500855614	175	VISp	Cux2- CreERT2	Ai93(TITL- GCaMP6f)
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1	500061511	175	VISp	Cux2-	Ai93(TITL-
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1	500304314	175	ViSp	CreERT2	GCaMP6f)
2	501271265	175	ViSp	Cux2- CreERT2	Ai93(TITL- GCaMP6f)

```
# I keep NWB downloads off by default because a full Visual Coding cache is
# To actually download the selected files from this notebook, set cfg["allen
downloaded = download_selected_nwb_files(boc, experiments, download=bool(cfg
print("Downloaded experiment IDs:", downloaded)
```

Downloaded experiment IDs: []

```
import matplotlib.pyplot as plt

cre_depth_summary = (
    experiments
    .groupby(["cre_line", "imaging_depth", "putative_layer"])
    .size()
    .reset_index(name="n_experiments")
    .sort_values(["putative_layer", "cre_line", "imaging_depth"])
)

display(cre_depth_summary)

save_table(
    cre_depth_summary,
    paths.tables_dir / "01_selected_experiments_by_cre_depth_layer.csv"
)

fig, ax = plt.subplots(figsize=(8, 4))

labels = (
    cre_depth_summary["cre_line"].astype(str)
    + "\n"
    + cre_depth_summary["putative_layer"].astype(str)
    + ", "
    + cre_depth_summary["imaging_depth"].astype(str)
    + "  $\mu$ m"
)

ax.bar(labels, cre_depth_summary["n_experiments"])

ax.set_xlabel("Cre line / layer / imaging depth")
ax.set_ylabel("Selected experiments")
ax.set_title("Selected Allen V1 natural movie experiments")
ax.tick_params(axis="x", rotation=45)

save_figure(fig, paths.figures_dir / "01_selected_experiments_by_cre_depth_1
plt.show()
```

	cre_line	imaging_depth	putative_layer	n_experiments
0	Cux2-CreERT2	175	L2/3	3



